

BU MET CS544 A2 S2024 Exam Memorization Sheet

The topics to the left are the content/questions you should be able to describe/answer from memory (except for the formulas on the formula sheet). Complete the table by filling in the information, using your notes when you have to.

What are the three measures of center?	Mean, median, mode
What are the four measures of spread?	Standard deviation, IQR, range, variance
When will the standard deviation be zero?	When all values are the same
Interpret standard deviation	Average distance from the mean
Right Skewed vs Left Skewed	Right skewed, long tail to the right, mean to the right of median Left skewed, long tail to the left, mean to the left of median
Outlier Test	$1.5 \times \text{IQR}$ >3 STDEV away from mean
Describe a Distribution	Huh
What are the summary statistics that are resistant to outliers?	Median, IQR
What are the summary statistics that are not resistant to outliers?	Mean, variance, range, std dev
percentiles	Xth percentile = X% of values less than selected value

z-score	Standardized score for normal distributions
Standard Normal Distribution	Mean = 0, std dev = 1
What do the numbers 68-95-99.7 mean?	1, 2, and 3 std dev away from mean
What is the main difference between a segmented bar graph and a mosaic plot?	dafuq is a mosaic plot
Correlation Coefficient	y-residuals?
What three things does the value of r communicate?	amount of correlation, direction of correlation, ???
Coefficient of Determination	huh?
Describe a Scatterplot	wtf?
Residual	expected - actual
What does the LSRL minimize?	minimizes y-residuals
Interpret the y-intercept	value when $x = 0$
Interpret the slope	y/x

What are the 4 sources of bias in a study?	
What are 2 biased sampling methods?	convenience sampling, voluntary response
What is the difference between a cluster sample and a stratified sample?	stratified ensures equal number of different classes cluster allows for nearby options
Why do we block in experiments?	remove bias
Matched Pairs Experiment	both treatments given to same individual or equals in different groups
Why do we randomize in experiments?	remove biases
What is the main difference between an experiment and an observational study?	experiment has a test group and a control group tries to influence outcome in some way
What are the four principles of experimental design?	uhm random, double blind, ???, ???
What is the scope of inference for experiments and studies?	The population represented by the sample
What does it mean when two events are mutually exclusive?	Only one can happen
General Addition Rule	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
General Multiplication Rule	$P(A)P(B) = P(A B)P(B)$

What does it mean when two events are independent?	$P(A)P(B) = P(A \text{ and } B)$
How can you test if two events are independent? (two ways)	Calculate if $P(A B)P(B) = P(A \text{ and } B)$ think really hard
Expected value of a discrete random variable	Sum of $P(X = x) * x$ for all X
Standard deviation of a discrete random variable	Sheeeeeit
Adding and subtracting random variables	isn't this on the formula sheet?
Linear transformation of a random variable	Mean multiply by constant, std dev multiply by sqrt constant
What are the four conditions for a binomial setting?	Random, independent, two outcomes, set number of trials
Binomial Probabilities	$P(X = x) = p^x (1-p)^{(n-x)}$
Mean and standard deviation of a binomial random variable	I forgot
What are the four conditions for a geometric setting?	Random, independent, two outcomes, go until first success
Geometric Probabilities	$P(X = x) = (1-p)^{(x-1)} * p$

Mean and standard deviation of a geometric random variable	mean = $1/p$? std dev = no thanks
Sampling distribution of a sample proportion	mean of sampling distribution equals sample proportion std dev of sampling distribution = not sure
Sampling distribution of a sample mean	mean of sampling distribution equals sample mean std dev of sampling distribution = std dev of population std dev over sqrt sample size
1-proportion z interval	
1 sample t-interval	
1-proportion z test	
1 sample t test	
When do we use t instead of z?	
Interpret p-value	
What is a Type I Error?	
What is a Type II Error?	

What is power?	
P(Type I Error)	
What happens to your margin of error as your confidence increases?	
What happens to your margin of error as your sample size increases?	
What happens to Type I error, Type II error, and power as α decreases?	
How do you find the expected value on a two-way table for a chi-square test?	
How do you find the degrees of freedom in a two-way table?	
What is the main difference between a chi-square test for homogeneity and a chi-square test for association/independence?	
What are the conditions needed for inference about a slope?	
t-interval for β	
t-test for β	

